Fine-Tuning Learning Large Language Models (Transfer Learning)

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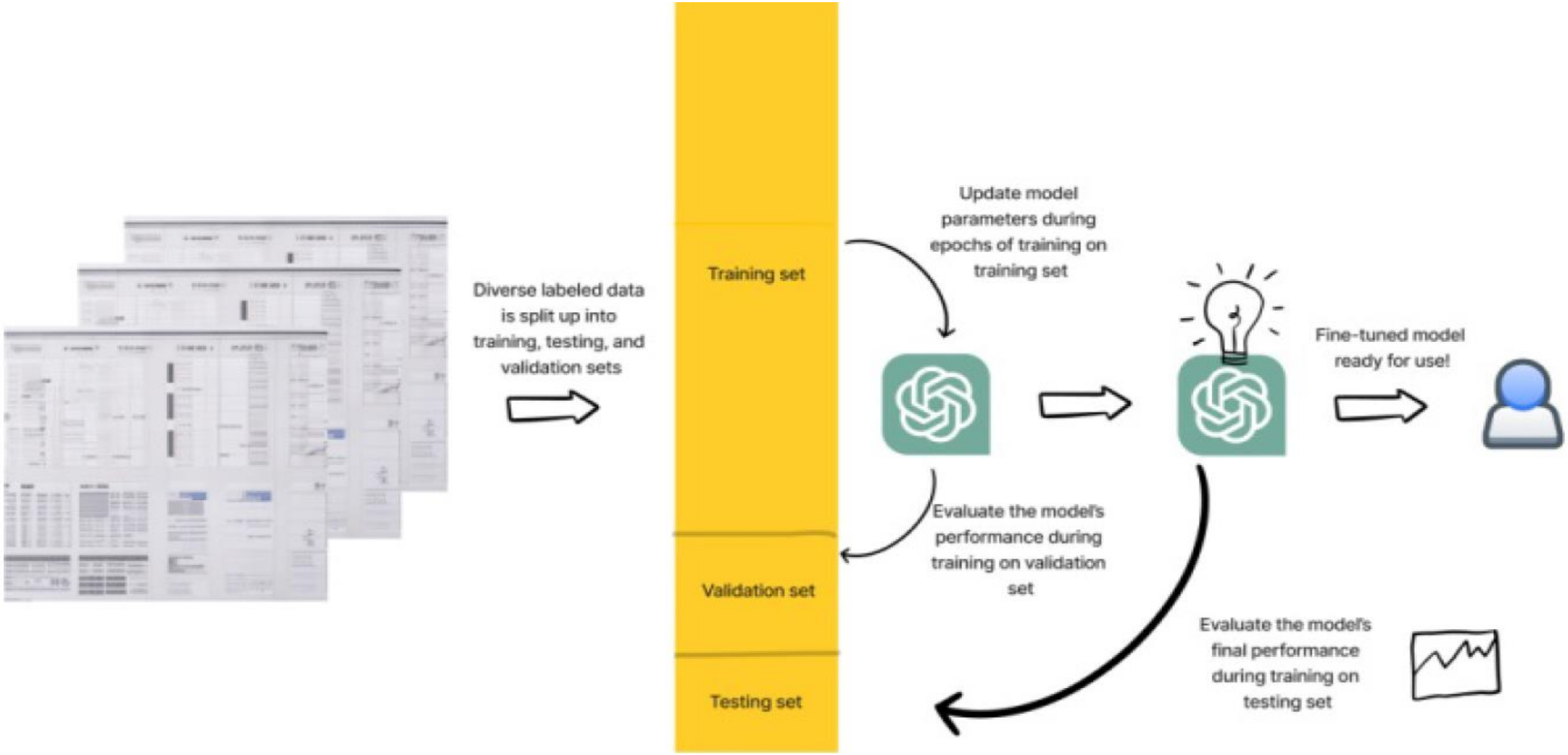
Transfer Learning

- In-context Learning like hard prompts and RAG.

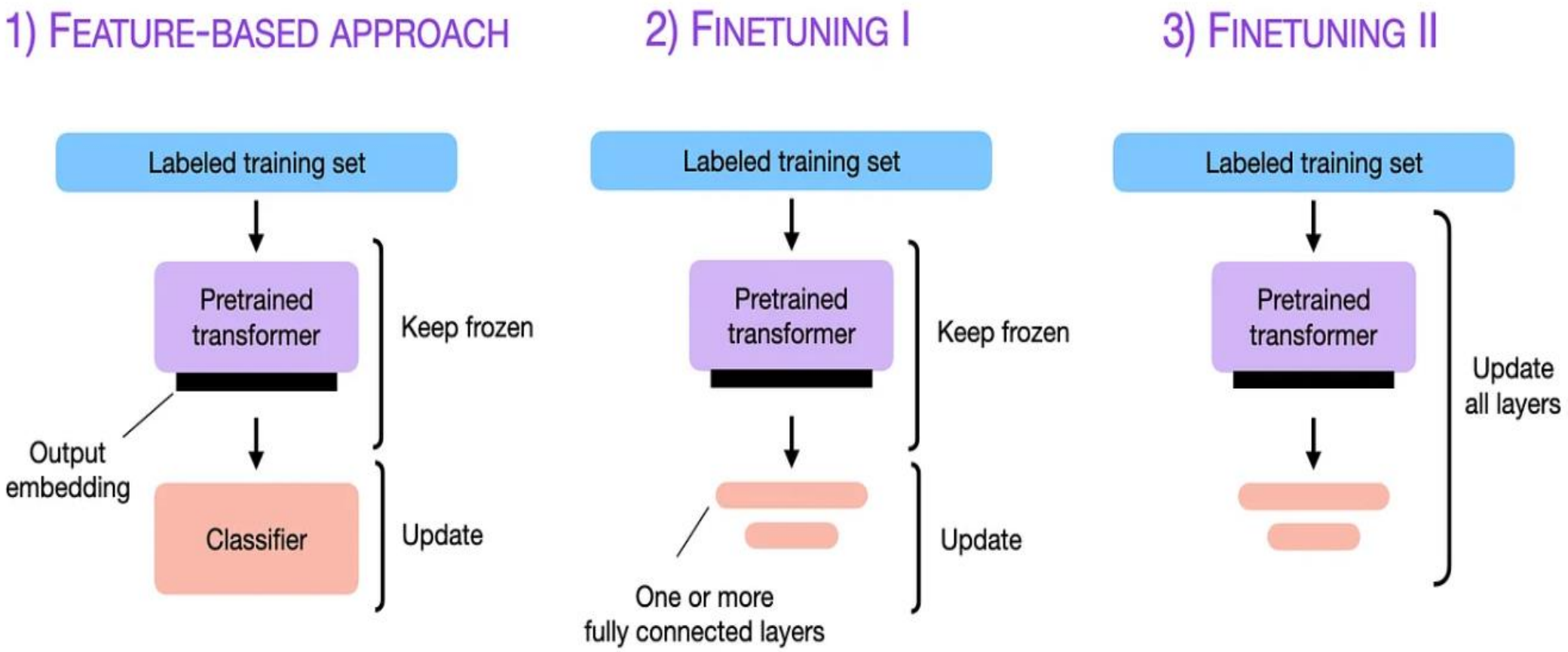
- Fine-tuning as a transfer learning approach.

	GPT-3 In-Context	PLM Fine-Tuning
Task-Agnostic	✔	✘
No Training	✔	✘
No External Costs	✘	✔
Scalable	✘	✔

The Fine-Tuning Process

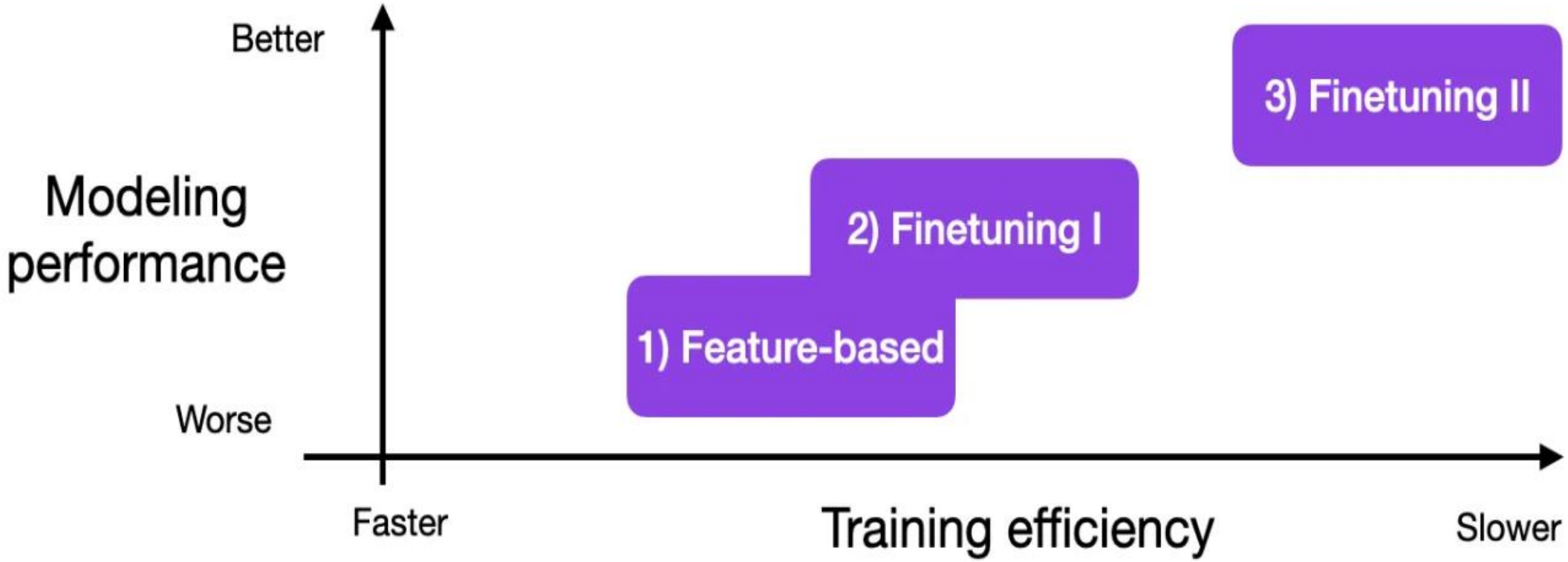


Fine-Tuning Approaches



https://github.com/rasbt/LLM-finetuning-scripts/tree/main/conventional/distilbert-movie-review

Computational and Modelling Performance Trade-offs



Closed-Source OpenAI PLM

- **Guidelines and Best Practices for Data:**
  - Data Quality: Ensure that the data used for fine-tuning is of high quality.
  - Data Diversity: Make sure your dataset is diverse, covering a broad range of scenarios.
  - Data Balancing: Maintaining a balanced distribution of examples across different tasks and domains.
  - Data Quantity: The total amount of data used to fine-tune the model.

- **Data Pre-processing:**
  - Removing duplicates
  - Splitting the data
  - Shuffling the training data
  - Creating the OpenAI JSONL format

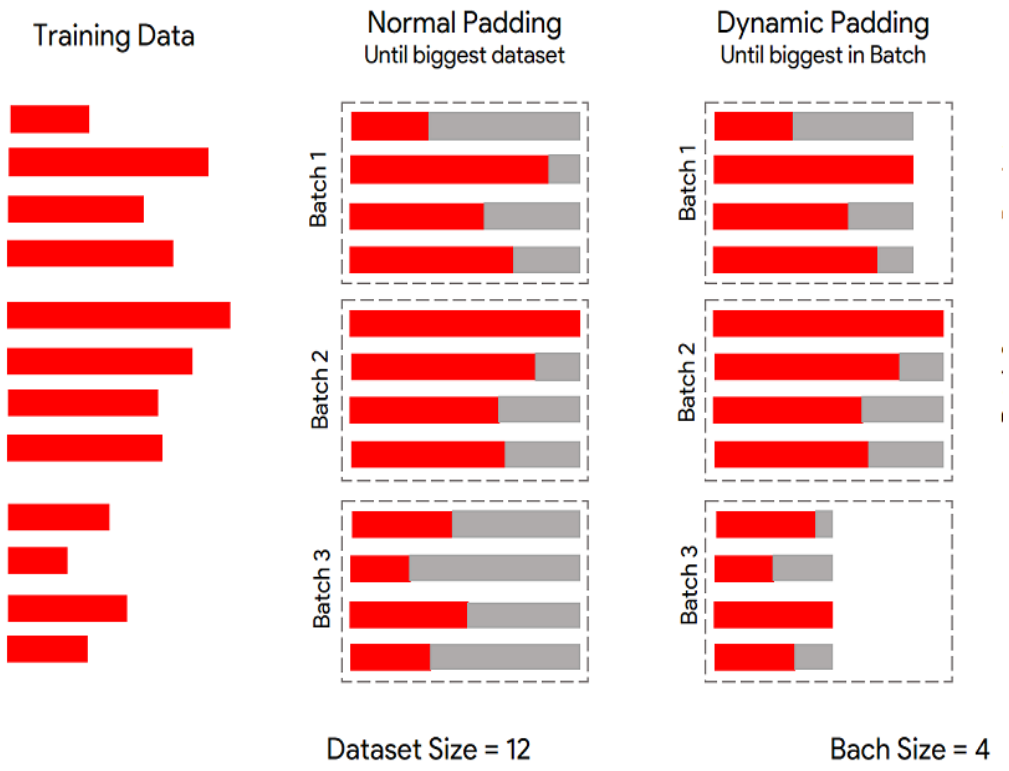
Integrating Fine-Tuned Models into Applications

- Identify your fine-tuned model:
  - For instance: "ft:babbage-002:personal::8EJokS4B"
- Use the OpenAI API
  - Use your OpenAI API to make requests to your fine-tuned model.
- Adapt any application logic
  - fine-tuned models may require different prompt structures or generate different output formats.
- Monitor and evaluate performance
  - Continuously monitor your fine-tuned model‘s performance and collect user feedback.

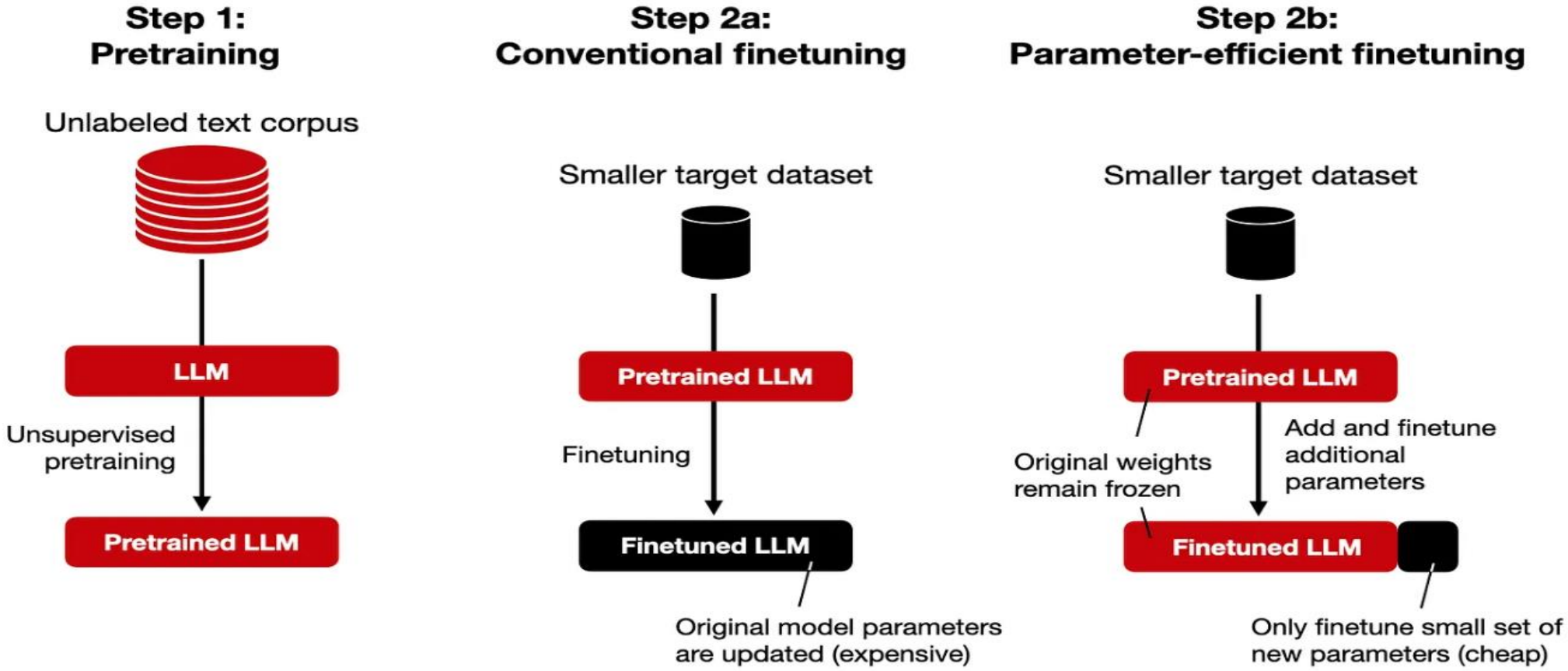
Fine-Tuning Open-Source LLMs

- **A Simple Fine-tuning Loop:**
  - Dataset
  - Data Collator
  - TrainingArguments
  - Weights & Biases and Trainer
- **General Tips:**
  - Adjusting Batch sizes and Gradient Accumulation
  - Dynamic Padding
  - Model Freezing

Dynamic Padding

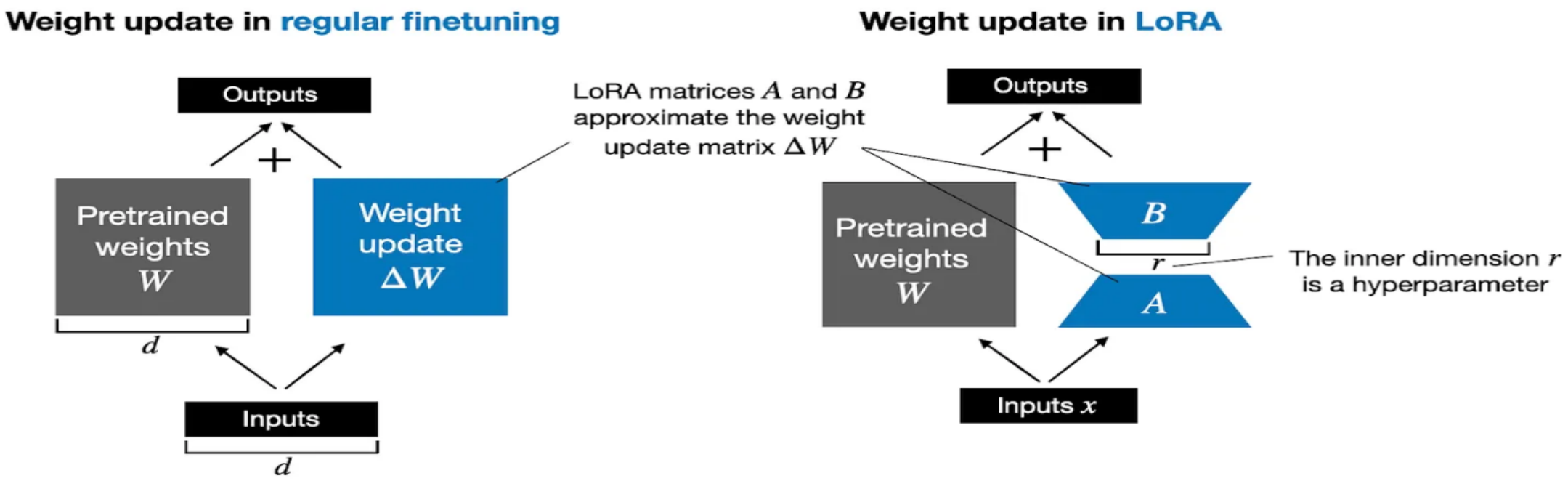


Parameter-Efficient Fine-Tuning (PEFT)



The idea of parameter-efficient fine-tuning techniques (like prompt-tuning, prefix-tuning, and adapter-tuning) is to add a small set of parameters to a pre-trained LLM. Only the newly added parameters are fine-tuned while all the parameters of the pre-trained LLM remain frozen.

Fine-tuning LLMs Using LoRA (Low-Rank Adaptation)



How much memory does this save? It depends on the rank r, which is a hyper-parameter. For example, if ΔW has 10,000 rows and 20,000 columns, it stores 200M parameters. If we choose A and B with r=8, then A has 10,000 rows and 8 columns, and B has 8 rows and 20,000 columns, that's 10,000×8 + 8×20,000 = 240,000 parameters, which is about 830× less than 200,000,000.

Tuning Language Models Approaches

- **Qlora:** Dettmers, T., Pagnoni, A., Holtzman, A. and Zettlemoyer, L., (2023). Qlora: Efficient finetuning of quantized llms. *arXiv preprint arXiv:2305.14314*.
- **Lora:** Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L. and Chen, W., (2021). Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*.
- **Prefix Tuning:** Li, X.L. and Liang, P., (2021). Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*.
- **Proxy Tuning:** Liu, A., Han, X., Wang, Y., Tsvetkov, Y., Choi, Y. and Smith, N.A., (2024). Tuning language models by proxy. *arXiv preprint arXiv:2401.08565*.
- **PEFT:** Xu, L., Xie, H., Qin, S.Z.J., Tao, X. and Wang, F.L., (2023). Parameter-efficient fine-tuning methods for pretrained language models: A critical review and assessment. *arXiv preprint arXiv:2312.12148*.

RAG Vs/And Fine-tuning

- **Key Considerations:**
  - External Knowledge
  - Labelled Training Dataset
  - Model Customisation
  - Suppress Hallucination
  - Scalability/Maintenance
  - Integration with Existing Systems
  - Cost

- **Use Cases:**
  - Summarisation
  - Question Answering Systems
  - Customer Support Automation/Chatbots

- Gupta, A., Shirgaonkar, A., Balaguer, A.D.L., Silva, B., Holstein, D., Li, D., Marsman, J., Nunes, L.O., Rouzbahman, M., Sharp, M. and Mecklenburg, N., (2024) RAG vs Fine-tuning: Pipelines, Tradeoffs, and a Case Study on Agriculture. *arXiv preprint arXiv:2401.08406*.

ngiyabonga
danke
謝謝
teşekkür ederim
спасибо
dank je
thank you
gracias
tapadh leat
bedankt
mauruuru
hvala
muchchakkeram
dziękuje
go raibh maith agat
sukriya
kop khun krap
taiku
obrigado
sagolun
arigato
tak
dakujem
grazie
terima kasih
мерси
감사합니다
merci
شكراً